

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)	I	{Section/ part} of a {DNA (molecule) / chromosome} which {determines/ codes for} {characteristics/ traits/ phenotype/ protein/ sequence of amino acids}	1			1		
			II	Change in a {gene/ DNA/ chromosome/ genetic make-up/ triplet code}	1			1		
		(ii)		{DNA / genetic} profiling/ {gene/ genome} sequencing	1			1		
	(b)	(i)		(leu) – cys – ser (leu) – tyr – ser (1) (1) Mark as columns			2	2		
		(ii)		• {TGT / cysteine/ cys} changed to {TAT / tyrosine/ tyr}/ {allele/ triplet code/ order of bases} changed (1) • giving {different/ wrong} amino acid/ {different/ wrong} sequence of amino acids (1) • iron regulating protein {does not work/ not produced}/ {different/ wrong} protein is produced/ protein with different function is produced/ protein is changed (1)		3		3		
	(c)	(i)		Phenotype: {carrier/ unaffected/ normal iron/ no haemochromatosis} x {affected/ haemochromatosis/ more iron/ sufferer/ has condition} (1) Genotype: Hh x hh (1) Correct gametes (1) Correct mechanics (1)		1 1 1 1		4	2	1

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		(ii)		50%/ 1/2/ 0.5 correct from Punnett square in (i)		1		1	1	
	(d)			A. (Mutation) has led to variation (1) B. {Mutant allele/ mutation} gives an {advantage/ better chance of survival}/ {people with haemochromatosis/ the (genetic) condition / description of haemochromatosis} have a better chance of surviving/ ORA (1) C. Because individuals can take in more iron (from diet) / ORA (1) D. (They reproduced and) <u>passed</u> (HFE) {allele/ gene/ mutation} on to next generation (1)	1 1	1 1		4		
				Total mark for question 8	5	10	2	17	3	1